



Uterine Sarcoma Treatment (PDQ®)–Health Professional Version

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General Information About Uterine Sarcoma

Uterine sarcoma accounts for less than 1% of gynecologic malignancies and 2% to 5% of all uterine malignancies.^[1] The following tumors arise primarily from three distinct tissues:

1. Carcinosarcoma arises in the endometrium, in other organs of mullerian origin, and accounts for 40% to 50% of all uterine sarcomas.
2. Leiomyosarcoma arises from myometrial muscle, with a peak incidence occurring at age 50 years, and accounts for 30% of all uterine sarcomas.
3. Sarcoma arises in the endometrial stroma, with a peak incidence occurring before menopause for the low-grade tumors and after menopause for the high-grade tumors accounts for 15% of all uterine sarcomas.

Questions?

These three distinct entities are often grouped together as uterine sarcomas; however, each type of tumor is being studied in separate clinical trials.

Carcinosarcoma (the preferred designation by the World Health Organization [WHO]) is also referred to as mixed mesodermal sarcoma or mullerian tumor. Controversy exists about the following issues:

- Whether carcinosarcoma is a true sarcoma.
- Whether the sarcomatous elements are actually derived from a common epithelial-cell precursor that also gives rise to the usually more abundant adenocarcinomatous elements.

The stromal components of carcinosarcoma are further characterized by homologous elements, such as malignant mesenchymal tissue considered possibly native to the uterus, or heterologous elements, such as striated muscle, cartilage, or bone, which are foreign to the uterus. Carcinosarcoma parallels endometrial cancer in its postmenopausal predominance and in other epidemiological features. Increasingly, the treatment of carcinosarcoma is becoming similar to combined modality approaches for endometrial adenocarcinoma.

Other rare forms of uterine sarcoma also fall under the WHO classification for mesenchymal and mixed tumors of the uterus. These sarcomas include the following:[2,3]

- Mixed endometrial stromal and smooth muscle tumors.
- Adenosarcomas, in which the epithelial elements appear benign within a malignant mesenchymal background.
- Embryonal botryoides or rhabdomyosarcomas, which are found almost exclusively in infants.
- PEComa—a perivascular epithelial-cell tumor that may behave in a malignant fashion, which is the latest to be added.

For more information, see [Childhood Rhabdomyosarcoma Treatment](#).

Risk Factors

The only documented etiological factor in 10% to 25% of these malignancies is prior pelvic radiation therapy, which is often administered for benign uterine bleeding that began 5 to 25 years earlier. An increased incidence of uterine sarcoma has been associated with tamoxifen in the treatment of breast cancer. Subsequently, increases have also been noted when tamoxifen was given to prevent breast cancer in women at increased risk—a possible result of the estrogenic effect of tamoxifen on the uterus. Because of this increase, patients taking tamoxifen should have follow-up pelvic examinations and should undergo endometrial biopsy if there is any abnormal uterine bleeding.[4-6]

Prognosis

The prognosis for women with uterine sarcoma primarily depends on the extent of disease at the time of diagnosis.[7] For women with carcinosarcoma, significant predictors of metastatic disease at initial surgery include:[7]

- Isthmic or cervical location.
- Lymphatic vascular space invasion.
- Serous and clear cell histology.
- Grade 2 or 3 carcinoma.

These factors, in addition to the following ones, correlate with a progression-free interval:[7]

- Adnexal spread.
- Lymph node metastases.
- Tumor size.

- Peritoneal cytologic findings.
- Depth of myometrial invasion.

Factors that bear no relationship to the presence or absence of metastases at surgical exploration include:

- Presence or absence of stromal heterologous elements.
- Types of such elements.
- Grade of the stromal components.
- Mitotic activity of the stromal components.

In one study, women with well-differentiated sarcomatous components or carcinosarcomas had significantly longer progression-free intervals than those with moderately to poorly differentiated sarcomas for the homologous and heterologous types. The recurrence rate was 44% for homologous tumors and 63% for heterologous tumors. The type of heterologous sarcoma had no effect on the progression-free interval.

For women with leiomyosarcomas, some investigators consider tumor size to be the most important prognostic factor. Women with tumors larger than 5.0 cm in maximum diameter have a poor prognosis.[8] However, in a Gynecologic Oncology Group study, the mitotic index was the only factor significantly related to progression-free interval.[7] Leiomyosarcomas matched for other known prognostic factors may be more aggressive than their carcinosarcoma counterparts.[9] The 5-year survival rate for women with stage I disease, which is confined to the corpus, is approximately 50% versus 0% to 20% for the remaining stages.

Surgery alone can be curative if the malignancy is contained within the uterus. The value of pelvic radiation therapy is not established. Current studies consist primarily of phase II chemotherapy trials for patients with advanced disease. Adjuvant chemotherapy following complete resection for patients with stage I or II disease was not found to be effective in a randomized trial.[10] Yet, nonrandomized trials have reported improved survival following adjuvant chemotherapy with or without radiation therapy.[11-13]

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Cellular Classification of Uterine Sarcoma

The most common histological types of uterine sarcomas include:

- Carcinosarcoma (mixed mesodermal sarcoma [40%–50%]).

- Leiomyosarcoma (30%).
- Endometrial stromal sarcoma (15%).

The uterine neoplasm classification of the International Society of Gynecologic Pathologists and the World Health Organization uses the term **carcinosarcoma** for all primary uterine neoplasms containing malignant elements of both epithelial and stromal light microscopic appearances, regardless of whether malignant heterologous elements are present.^[1]

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1. Silverberg SG, Major FJ, Blessing JA, et al.: Carcinosarcoma (malignant mixed mesodermal tumor) of the uterus. A Gynecologic Oncology Group pathologic study of 203 cases. Int J Gynecol Pathol 9 (1): 1-19, 1990. [\[PUBMED Abstract\]](#)

Stage Information for Uterine Sarcoma

FIGO Staging

The Fédération Internationale de Gynécologie et d’Obstétrique (FIGO) and the American Joint Committee on Cancer have designated staging to define uterine sarcoma; the FIGO system is most commonly used.^[1,2]

The FIGO staging system has two divisions, one for leiomyosarcoma and endometrial stromal sarcoma, and one for adenosarcoma. Carcinosarcoma is staged using the designated endometrial carcinoma definitions. For more information, see [Endometrial Cancer Treatment](#).^[1]

Table 1. FIGO Definitions of Uterine Sarcoma–Leiomyosarcoma and Endometrial Stromal Sarcoma^a

Stage	Definition
I	Tumor limited to uterus.
FIGO = Fédération Internationale de Gynécologie et d’Obstétrique.	
^a Adapted from the Fédération Internationale de Gynécologie et d’Obstétrique. ^[1]	

Stage	Definition
–IA	Tumor ≤5 cm.
–IB	Tumor >5 cm.
II	Tumor extends beyond the uterus, within the pelvis.
–IIA	Adnexal involvement.
–IIB	Involvement of other pelvic tissues.
III	Tumor invades abdominal tissues (not just protruding into the abdomen).
–IIIA	One site.
–IIIB	More than one site.
–IIIC	Metastasis to pelvic and/or para-aortic lymph nodes.
IV	
–IVA	Tumor invades bladder and/or rectum.
–IVB	Distant metastasis.
FIGO = Fédération Internationale de Gynécologie et d’Obstétrique.	
ªAdapted from the Fédération Internationale de Gynécologie et d’Obstétrique.[1]	

Table 2. FIGO Definitions of Uterine Sarcoma–Adenosarcomaª

Stage	Definition
I	Tumor limited to uterus.
–IA	Tumor limited to endometrium/endocervix with no myometrial invasion.
–IB	Less than or equal to half myometrial invasion.
–IC	More than half myometrial invasion.
II	Tumor extends to the pelvis.
–IIA	Adnexal involvement.
–IIB	Tumor extends to extrauterine pelvic tissue.
III	Tumor invades abdominal tissues (not just protruding into the abdomen).
–IIIA	One site.
–IIIB	More than one site.
–IIIC	Metastasis to pelvic and/or para-aortic lymph nodes.
IV	
–IVA	Tumor invades bladder and/or rectum.
–IVB	Distant metastasis.
FIGO = Fédération Internationale de Gynécologie et d'Obstétrique.	
^a Adapted from the Fédération Internationale de Gynécologie et d'Obstétrique.[1]	

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Treatment Option Overview for Uterine Sarcoma

Surgery is often the principal means of diagnosis and is the primary treatment for all patients with uterine sarcoma. If the diagnosis is known, the extent of surgery is planned according to the stage of the tumor. Hysterectomy is usually performed when a uterine malignancy is suspected, except for rare instances when preservation of the uterus in a young patient is deemed safe for the type of cancer (e.g., a totally confined low-grade leiomyosarcoma in a woman who desires to retain childbearing potential). Medically suitable patients with the preoperative diagnosis of uterine sarcoma are considered candidates for abdominal hysterectomy, bilateral salpingo-oophorectomy, and pelvic and periaortic selective lymphadenectomy. Cytologic washings are obtained from the pelvis and abdomen. Thorough examination of the diaphragm, omentum, and upper abdomen is performed.

There is no firm evidence from a prospective study that adjuvant chemotherapy or radiation therapy is of benefit for patients with uterine sarcoma.[1] In one Gynecologic Oncology Group (GOG) study, the use of adjuvant doxorubicin did not alter the survival rate of patients with resected stage I or stage II uterine sarcomas; however, interpretation of these results is difficult because this study included some patients who received radiation and three types of uterine sarcomas that have variable responses to doxorubicin.[1][[Level of evidence A1](#)] Because the risk of disease recurrence is high, even with localized presentations, many physicians have considered the use of adjuvant chemotherapy or radiation therapy.[2] Another GOG study ([GOG-0150](#) [NCT00002546]) addressed radiation therapy versus adjuvant chemotherapy.[3]

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Treatment of Stage I Uterine Sarcoma

Treatment Options for Stage I Uterine Sarcoma

Treatment options for stage I uterine sarcoma include:

1. Surgery (total abdominal hysterectomy, bilateral salpingo-oophorectomy, and pelvic and periaortic selective lymphadenectomy).
2. Surgery plus pelvic radiation therapy.
3. Surgery plus adjuvant chemotherapy.
4. Surgery plus adjuvant radiation therapy.

A nonrandomized Gynecologic Oncology Group study examined the effect of pelvic radiation therapy on patients with stage I and II carcinosarcomas. Patients who had pelvic radiation therapy had a significant reduction in tumor recurrences within the radiation treatment field but no alteration in survival.[1] A large nonrandomized study demonstrated improved survival and a lower local failure rate in patients with mixed mullerian tumors following postoperative external and intracavitary radiation therapy.[2] One nonrandomized study that predominantly included patients with carcinosarcomas showed that adjuvant chemotherapy with cisplatin and doxorubicin benefitted participants.[3]

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Treatment of Stage II Uterine Sarcoma

Treatment Options for Stage II Uterine Sarcoma

Treatment options for stage II uterine sarcoma include:

1. Surgery (total abdominal hysterectomy, bilateral salpingo-oophorectomy, and pelvic and periaortic selective lymphadenectomy).
2. Surgery plus pelvic radiation therapy.
3. Surgery plus adjuvant chemotherapy.
4. Surgery plus adjuvant radiation therapy.

A nonrandomized Gynecologic Oncology Group study examined the effect of pelvic radiation therapy on patients with stage I and II carcinosarcomas. Patients who had pelvic radiation therapy had a significant reduction in tumor recurrences within the radiation treatment field but no alteration in survival.[1] One nonrandomized study that predominantly included patients with carcinosarcomas showed that adjuvant chemotherapy with cisplatin and doxorubicin benefitted participants.[2]

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Treatment of Stage III Uterine Sarcoma

Treatment Options for Stage III Uterine Sarcoma

Treatment options for stage III uterine sarcoma include:

1. Surgery (total abdominal hysterectomy, bilateral salpingo-oophorectomy, pelvic and periaortic selective lymphadenectomy, and resection of all gross tumor).
2. Surgery plus pelvic radiation therapy (under clinical evaluation).
3. Surgery plus adjuvant chemotherapy (under clinical evaluation).

Phase II chemotherapy studies by the Gynecologic Oncology Group have documented some antitumor activity for cisplatin, doxorubicin, and ifosfamide.[1,2] These studies have also documented differences in response leading to separate trials for patients with carcinosarcomas and leiomyosarcomas. In patients previously untreated with chemotherapy, ifosfamide had a 32.2% response rate in patients with carcinosarcomas [3] and a 17.2% partial response rate in patients with leiomyosarcomas.[2]

GOG-108 was a randomized trial that examined the use of ifosfamide with or without cisplatin as first-line therapy for patients with measurable advanced or recurrent carcinosarcomas. Patients in the combination arm had a higher response rate (54% vs. 34%) and longer progression-free survival (PFS) (6 months vs. 4 months). However, patients did not have a significant improvement in survival (9 months vs. 8 months).[4][[Level of evidence A1](#)] The follow-up [GOG-0161](#) study [NCT00003128] used 3-day ifosfamide regimens (instead of the more toxic 5-day regimen in the preceding study) given alone or in combination with paclitaxel (with filgrastim starting on day 4).[5] The combination was superior in response rates (45% vs. 29%), PFS (8.4 months vs. 5.8 months), and overall survival (13.5 months vs. 8.4 months). The hazard ratio for death favored the combination (0.69; 95% confidence interval, 0.49–0.97).[5] [[Level of evidence A1](#)] In this study, 52% of 179 evaluable patients had recurrent disease, 18% had stage III disease, and 30% had stage IV disease. In addition, imbalances were present in the sites of disease and in the use of prior radiation therapy, and 30 patients were excluded for wrong pathology.

A role for chemotherapy as an adjuvant to surgery has not been established.

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Treatment of Stage IV Uterine Sarcoma

Treatment Options for Stage IV Uterine Sarcoma

There is currently no standard therapy for patients with stage IV disease. These patients should enroll in an ongoing clinical trial.

Phase II chemotherapy studies by the Gynecologic Oncology Group have documented some antitumor activity for cisplatin, doxorubicin, and ifosfamide.[1,2] These studies have also documented differences in response leading to separate trials for patients with carcinosarcomas and leiomyosarcomas. In patients previously untreated with chemotherapy, ifosfamide had a 32.2% response rate in patients with carcinosarcomas,[2] a 33% response rate in patients with endometrial stromal cell sarcomas,[3], and a 17.2% partial response rate in patients with leiomyosarcomas.[4] Doxorubicin in combination with dacarbazine or cyclophosphamide is no more active than doxorubicin alone for advanced disease.[5,6] Cisplatin has activity as first-line therapy and minimal activity as second-line therapy for patients with carcinosarcomas, but cisplatin is inactive as first- or second-line therapy for patients with leiomyosarcomas.[1,7]

GOG-108 was a randomized trial that examined the use of ifosfamide with or without cisplatin as first-line therapy for patients with measurable advanced or recurrent carcinosarcomas. Patients in the combination arm had a higher response rate (54% vs. 34%) and longer progression-free survival (PFS) (6 months vs. 4 months). However, patients did not have a significant improvement in survival (9 months vs. 8 months).[8][[Level of evidence A1](#)] The follow-up [GOG-0161](#) study [NCT00003128] used 3-day ifosfamide regimens (instead of the more toxic 5-day regimen in the preceding study) given alone or in combination with paclitaxel (with filgrastim starting on day 4).[9] The combination was superior in response rates (45% vs.

29%), PFS (8.4 months vs. 5.8 months), and overall survival (13.5 months vs. 8.4 months). The hazard ratio for death favored the combination (0.69; 95% confidence interval, 0.49–0.97).[9] [Level of evidence A1] In this study, 52% of 179 evaluable patients had recurrent disease, 18% had stage III disease, and 30% had stage IV disease. In addition, imbalances were present in the sites of disease and in the use of prior radiation therapy, and 30 patients were excluded for wrong pathology.

Current Clinical Trials

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Treatment of Recurrent Uterine Sarcoma

Treatment Options for Recurrent Uterine Sarcoma

There is currently no standard therapy for patients with recurrent disease. These patients should enroll in an ongoing clinical trial.

Phase II chemotherapy studies by the Gynecologic Oncology Group have documented some antitumor activity for cisplatin, doxorubicin, and ifosfamide.[\[1,2\]](#) These studies have also documented differences in response leading to separate trials for patients with carcinosarcomas and leiomyosarcomas. In patients previously untreated with chemotherapy, ifosfamide had a 32.2% response rate in patients with carcinosarcomas,[\[2\]](#) a 33% response rate in patients with endometrial stromal cell sarcomas,[\[3\]](#) and a 17.2% partial response rate in patients with leiomyosarcomas.[\[4\]](#) Doxorubicin in combination with dacarbazine or cyclophosphamide is no more active than doxorubicin alone for recurrent disease.[\[5,6\]](#) Cisplatin has activity as first-line therapy and minimal activity as second-line therapy for patients with carcinosarcomas, but cisplatin is inactive as first- or second-line therapy for patients with leiomyosarcomas.[\[1,7\]](#) A regimen of gemcitabine plus docetaxel had a 53% response rate in patients with unresectable leiomyosarcomas and is undergoing further study.[\[8\]](#)

GOG-108 was a randomized trial that examined the use of ifosfamide with or without cisplatin as first-line therapy for patients with measurable advanced or recurrent carcinosarcomas. Patients in the combination arm had a higher response rate (54% vs. 34%) and longer progression-free survival (PFS) (6 months vs. 4 months). However, patients did not have a significant improvement in survival (9 months vs. 8 months).[\[9\]](#)[\[Level of evidence A1\]](#) The follow-up [GOG-0161](#) study [[NCT00003128](#)] used 3-day ifosfamide regimens (instead of the more toxic 5-day regimen in the preceding study) given alone or in combination with paclitaxel (with filgrastim starting on day 4).[\[10\]](#) The combination was superior in response rates (45% vs. 29%), PFS (8.4 months vs. 5.8 months), and overall survival (13.5 months vs. 8.4 months). The hazard ratio for death favored the combination (0.69; 95% confidence interval, 0.49–0.97).[\[10\]](#) [\[Level of evidence A1\]](#) In this study, 52% of 179 evaluable patients had recurrent disease, 18% had stage III disease, and 30% had stage IV disease. In addition, imbalances were present in the sites of disease and in the use of prior radiation therapy, and 30 patients were excluded for wrong pathology.

Radiation therapy may be an effective method of palliative care for patients with carcinosarcoma who have localized recurrence in the pelvis confirmed by computed tomography. Phase I and II clinical trials are appropriate for patients who have disease recurrence with distant metastasis and are unresponsive to first-line phase II trials. High-dose progesterone hormone therapy may be of some benefit to patients with low-grade stromal sarcoma.^[11]

Current Clinical Trials

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Latest Updates to This Summary (12/17/2024)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Editorial changes were made to this summary.

This summary is written and maintained by the [PDQ Adult Treatment Editorial Board](#), which is editorially independent of NCI. The summary reflects an independent review of the literature and does not represent a policy statement of NCI or NIH. More information about summary policies and the role of the PDQ Editorial Boards in maintaining the PDQ summaries can be found on the [About This PDQ Summary](#) and [PDQ® Cancer Information for Health Professionals](#) pages.

About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the treatment of uterine sarcoma. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

Reviewers and Updates

This summary is reviewed regularly and updated as necessary by the [PDQ Adult Treatment Editorial Board](#), which is editorially independent of the National Cancer Institute (NCI). The

summary reflects an independent review of the literature and does not represent a policy statement of NCI or the National Institutes of Health (NIH).

Board members review recently published articles each month to determine whether an article should:

- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

Changes to the summaries are made through a consensus process in which Board members evaluate the strength of the evidence in the published articles and determine how the article should be included in the summary.

The lead reviewer for Uterine Sarcoma Treatment is:

- Marina Stasenکو, MD (New York University Medical Center)

Any comments or questions about the summary content should be submitted to Cancer.gov through the NCI website's [Email Us](#). Do not contact the individual Board Members with questions or comments about the summaries. Board members will not respond to individual inquiries.

Levels of Evidence

Some of the reference citations in this summary are accompanied by a level-of-evidence designation. These designations are intended to help readers assess the strength of the evidence supporting the use of specific interventions or approaches. The PDQ Adult Treatment Editorial Board uses a [formal evidence ranking system](#) in developing its level-of-evidence designations.

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